

Speak Like A Mathematician:

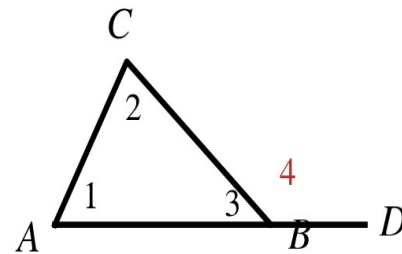
interior angle:

exterior angle:

remote interior angles:

corollary:

equiangular triangle:



Triangle Sum Theorem and the Exterior Angle Theorem

Triangle Sum Theorem: the measures of the three interior angles of a triangle add to _____.

Exterior Angle Theorem: the measure of an exterior angle equals the _____ of the measures of the two _____ angles.

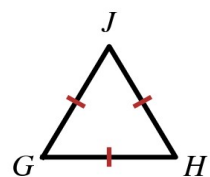
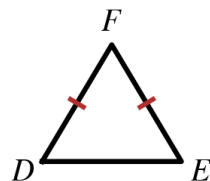
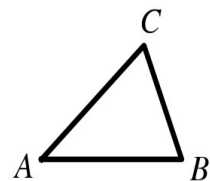
Corollary: the acute angles of a right triangle are _____. **Corollary:** each angle of an equiangular triangle measures _____.

Third Angles Theorem: if two angles of one triangle are congruent to two angles of another, then the _____ pair of angles is congruent.

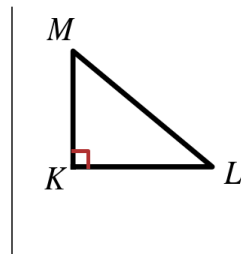
Example 1 - Classify Each Triangle

Classify each triangle at the right by its sides and by its angles.

- a) the triangle with vertices A, B, C
- b) the triangle with vertices D, E, F
- c) the triangle with vertices G, H, J
- d) the triangle with vertices K, L, M



SC Honors Geometry - Nappier
5.1 Angles of Triangles

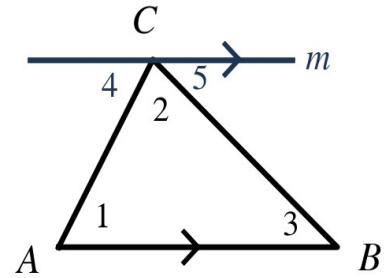


Example 2 - Prove the Triangle Sum Theorem

Given: $\triangle ABC$, with line m drawn through C parallel to $\overline{AB}$

Prove: $m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ$

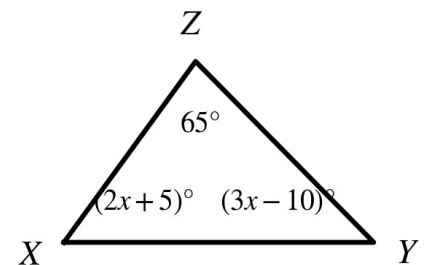
Statements	Reasons
1. $m \parallel \overline{AB}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.



Example 3 - Find the Angle Measures

Use the Triangle Sum Theorem to find x , then find each angle measure.

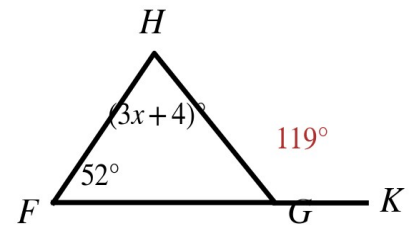
- Write an equation.
- Solve for x .
- Find $m\angle X$ and $m\angle Y$. Then check your work.



Example 4 - Use the Exterior Angle Theorem

In the diagram, $\angle HGK$ is an exterior angle of $\triangle FGH$.

- Name the two remote interior angles for $\angle HGK$.
- Write and solve an equation for x .
- Find $m\angle H$. Then find $m\angle FGH$ and check the Triangle Sum Theorem.



Example 5 - The Right Triangle Corollary

In $\triangle PQR$, $\angle P$ is a right angle.

- Why must the two acute angles be complementary?
- Write and solve an equation for y .
- Find both acute angle measures.

Example 6 - The Third Angles Theorem

In the diagram, $\angle A \cong \angle D$ and $\angle B \cong \angle E$.

- What does the Third Angles Theorem let you conclude?
- If $m\angle A = 47^\circ$ and $m\angle B = 68^\circ$, find $m\angle F$.

